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Health Priorities for Rural Alaska Native Communities at the Extremes of Weather and Climate Change

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Abstract

Background: Climate change is occurring at an accelerated pace in high-latitude regions, with implications for the health and well-being of rural northern communities--places that have long navigated significant health disparities with limited resources. Here, we present a community-driven effort to identify community priorities for improving health in the context of a changing climate.

Methods: We used Q-methodology in four rural Alaska Native communities to systematically identify complex perspectives related to climate change and health priorities. The mixed qualitative-quantitative approach helped to engage the community, prioritize the most locally relevant issues, and identify targets for future interventions.

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Results: Consistently top-ranked priorities include food security, Indigenous knowledge and practices, and the transmission of culture and traditional ways of living to the younger generations. Community perspectives reflect a holistic view of health that connects climate change to individual health and community well-being. These three components are linked together through the social-ecological landscape, providing resources that are embedded in the local and Indigenous cultural practices.

Conclusions: We show that a community-engaged approach can generate consensus data, directing research and interventions towards the highest needs and priorities of Alaska Native communities highly affected by climate change. Overall, participating rural Alaska Native communities prioritized the impacts of climate change on traditional ways of living from the land to subsist and thrive in these remote, culturally important areas. Additionally, the well-being and futures of young people arose as a primary shared concern, directing interventions to address the impacts of climate change on health towards place-based solutions that protect and preserve environmental resources and engage young people in the transmission of traditional knowledge and practices on the land.

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Keywords

Alaska, Climate, Health, Priorities, Indigenous, Community, Tradition, Subsistence.

Background

Climate change poses a threat to global health, with particularly acute impacts in arctic regions such as Alaska, which are warming at two to four times the rate of other regions (Rantanen et al. 2022). Alaska's unique environmental, cultural, and socioeconomic landscape necessitates a localized approach to understanding and

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addressing these challenges. In addition, due to the accelerated speed of warming at high latitudes, research in Alaska can serve as a model for the Arctic and other regions that experience slower change. It is important to identify Alaska Native (AN) community priorities regarding climate change and health to guide the development of targeted interventions and policy decisions that reflect local realities, priorities, and needs.

Health and well-being are not distributed equally among the residents of the US, and many factors influence health disparities between populations. Even the definitions of health and how to assess health and well-being can differ between and within communities (Lewis 2010; Kral et al. 2011; Donatuto et al. 2020). Indigenous peoples often hold understandings of health that extend beyond Western biomedical frameworks (Donatuto et al. 2016, 2020). To advance health equity for all Americans, we need population-specific measures that can capture the factors driving well-being within particular cultural contexts. Socio-economic, historical, and environmental factors interact with individual determinants of health to influence the well-being of individuals and communities in the context of physical, mental, and behavioral health. These factors also influence each other at different scales, creating a complex web of interactions and influences. In the circumpolar North, many of the drivers of health are unique from those at lower latitudes, resulting in exacerbated health disparities (Allen et al. 2014; Wexler et al. 2024).

Alaska is home to 229 federally recognized AN Tribes living across an ecologically diverse geography that is roughly one-fifth the size of the contiguous United States. Alaska today stands at the center of multiple, colliding environmental and health crises with impacts among AN people living in remote rural communities where health and

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well-being are exigently place-based (Wheat et al. 2022). Individual health outcomes in AN communities are socio-ecologically determined, being tied inextricably to the land, animals, and environment, an Indigenous conceptualization of health that aligns with the Western science-based One Health framework, which is based on the principle that human, animal, and environmental health are intrinsically linked and dependent on each other. This concept has special relevance in the circumpolar North (Hueffer et al. 2019). Over the last century, AN people have experienced rapid and imposed environmental and social changes because of the colonial incursions, marginalizing Indigenous communities and cultures (Crawford 2013). While the Arctic colonial encounter improved some physical health outcomes, it also created structures of inequity that ultimately removed Indigenous people as primary drivers of public health policy and as managers of natural resources (Rasmus, Allen, et al. 2014; Jernigan et al. 2023). Consequently, AN people experience serious health inequities such as disparities in suicide, substance use disorders, and cardiovascular disease (ANTHC 2021).

In addition to the historical, social, and economic drivers of health disparities, rapid environmental change further challenges traditional AN cultures and poses further threats to health and well-being that likely will disproportionately affect remote rural, largely Indigenous communities (Schwoerer et al. 2023). Principal among these environmental changes is ongoing climate change; the pace of warming is two- to fourfold or greater in the Arctic than the rest of the planet (Ruscio et al. 2015; Rantanen et al. 2022). Recent climate change impacts on the northern environment include melting sea ice, rising sea levels, increasing storm surges, flooding, salination of groundwater, ocean acidification, increased precipitation, thawing permafrost, less

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predictable river ice conditions, shifting vegetation patterns, changes in wildlife and fisheries populations, and more wildfires (Druckenmiller et al. 2025). In addition, these direct changes to the environment are predicted to also change infectious disease dynamics (Hueffer et al. 2013; National Academy of Sciences, Engineering, and Medicine 2020). While many of the specific consequences of a changing climate on infectious diseases, specifically zoonotic diseases, are still not very well understood, government agencies have started to set priorities and policies (Hueffer et al. 2011; Parkinson et al. 2014; National Academy of Sciences, Engineering, and Medicine 2020). It is, however, not clear how the priorities of the government agencies align with threats as perceived by local communities in Alaska.

The effects of climate change on health are potentially far-reaching and will likely have negative effects on Indigenous survivance and well-being in the North (Wilbur and Gone 2023). Despite the critical nature of these issues, there remain gaps in our understanding of (a) the specific priorities and concerns of AN communities regarding climate change and health, and (b) how to address these priorities in respectful and equitable ways to enhance well-being and climate security. Knowledge about community priorities is essential for developing effective and sustainable health interventions that are culturally appropriate and locally relevant. The objective of this study was to center Alaska Native communities in identifying and prioritizing climate-related health concerns.

Methods

Study settings

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This study was conducted in partnership with AN communities in two regions of Alaska, the Interior Alaska and Yukon–Kuskokwim (Western) regions of Alaska (See Figure 1), with two AN communities within each region. None of the four communities is connected to the road system. This isolation limits access to the types of goods and services found in cities, except for those that can be flown or barged in when weather permits. The remoteness of these communities poses economic challenges through high prices of store-bought products and limited opportunities for employment. Healthcare is also limited to acute care services available at most sites through trained community health workers. While all four communities share social and logistical features, the regions in which the communities are located are characterized by very different climates, geographies, ecologies, languages, and cultural traditions.



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Figure 1: Location of partner communities in Interior (Arctic Village and Tanana) and Western (Emmonak and Toksook Bay) Alaska.

Interior Alaska Region:

Interior Alaska is the traditional homeland of the Dene people and is characterized by a boreal forest ecosystem with warm summers and extreme cold during winter months. The upper and middle Yukon River and its tributaries form a connecting thread throughout the region. Many areas, especially in the northern Interior region, have large expanses of permafrost.

Arctic Village:

Vashrąįį K'ǫǫ (Arctic Village) is a Gwich'in community located at the northern edge of the Interior, in the southern foothills of the Brooks Range. The community consists of about 150 people (US Census 2020), who derive much of their food from the land by hunting caribou and moose, fishing for pike, grayling, whitefish, trout, and other local fish, and gathering various plants. The community is situated on the east fork of the Chandalar River, providing much of its fishing resources (Tanana Chiefs Conference 2026).

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Tanana:

The Koyukon village of Hohudodetlaatl Denh (Tanana) is located northwest of Fairbanks at the confluence of the Yukon and Tanana rivers. Hunting moose and other wildlife, as well as extensive salmon runs, have traditionally sustained this community of 246 people (US Census 2020). However, over the past decades, increasing restrictions on salmon fishing due to low returns have posed challenges for the community (Tanana Chiefs Conference 2026).

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Yukon-Kuskokwim Delta Region:

The Yukon-Kuskokwim Delta is the traditional homeland of the Yup'ik people in Western Alaska. It is characterized by a coastal tundra with few, if any, trees. The Kuskokwim and Yukon Rivers have traditionally provided ample fishing opportunities to sustain communities, while proximity to the Bering Sea also provides marine resources for traditional subsistence activities. However, here too, increasing restrictions on salmon fishing due to low returns have posed a persistent challenge.

Emmonak:

Imangaq (Emmonak) is located close to the mouth of the Yukon River and the Bering Sea. Traditionally, salmon fishing and subsistence hunting for marine and terrestrial resources have sustained the Yupik population of a little over 800 people (US Census 2020). For decades, robust salmon runs supported a local commercial fishing industry and processing plant. Recently, however, commercial fishing has been curtailed, practically eliminating this source of income. The community also serves as a subregional hub, providing services to several smaller villages in the region (City of Emmonak 2026).

Toksook Bay:

Nunakauyaq (Toksook Bay) is a Yup'ik community of around 600 people on Nelson Island along the Bering Sea coast (US Census 2020). Most of the population is engaged in fishing, marine mammal and musk ox hunting, as well as other subsistence activities. In Toksook Bay, residents of all ages are fluent in Yugtun, and many are monolingual or first-language speakers. The community provides mid-level health services to residents and surrounding communities as the subregional hub for Nelson Island (NOAA 2013).

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Partnership building

The foundation of our study is an alliance of researchers, communities, and agencies with a shared commitment to understanding and addressing the health impacts of climate change in AN communities. Through discussions that were centered on the initial priorities of community leaders and their feedback, we selected four rural, Indigenous communities in Alaska and co-developed methodologies to assess community priorities around the health impacts of climate change on these communities. In addition, we worked with the federally recognized Tribal governments in these communities to secure formal approval of our work and methodology. This deliberate community engagement process involved careful relationship building through weekly meetings of our leadership team that included a co-investigator from each community, community visits, and presentations to tribal governance organizations. This approach avoided preconceived ideas about what issues might be most relevant to communities because community members led the statement generation, rather than researchers who could instead be more focused on assumed priorities or their specific areas of expertise.

Prioritization of community concerns with Q-methodology

We used a modified application of the Q-methodology (Churruca et al. 2021; Sneegas et al. 2021; Hensel et al. 2022) to systematically identify and prioritize complex community perspectives on climate change and health. Q-methodology is a mixed qualitative-quantitative approach well suited to surfacing locally relevant concerns without imposing predetermined categories. The specific application of this methodology was shaped through extensive collaboration with our community partners and through

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discussions at two annual and three quarterly alliance meetings, ensuring the approach reflected community priorities from the outset. The Q-method helps to reveal social values, narratives, and perspectives related to complex social, environmental, and health issues (Ray 2011; Zabala et al. 2018; Moser, and Baulcomb 2020).

First, we facilitated focus-group discussions with community members to collect different localized perspectives on climate-related impacts on health and well-being. We engaged a total of 149 Community members (Arctic village: 17, Tanana: 43, Tooksok Bay: 42 Emmonak: 47). When key Indigenous knowledge holders in the community could not participate in focus groups, we conducted semi-structured interviews using the same prompts as in the focus groups. Notes from focus groups and interviews captured key opinions within the community and provided a holistic understanding of the connection between climate change and health. Individual statements were generated from discussion notes to complete the following sentence: “In my community, efforts on climate and health should focus on …”. Community members and researchers reviewed and revised the list of statements to reach a consensus. The final statements were constructed to enable consistent interpretation and ensure the final list of statements comprehensively addressed the issue with nuance but without being overly repetitive.

During the next step, individual community members sorted (e.g., Q-sort or rank) the statements from lowest to highest priority (e.g., least to most important) on a scale from 1 to 4 or 5, depending on the number of statements generated. We then analyzed the rankings by all participants within a community by generating the mean rank for each individual statement. This allowed us to estimate the highest to lowest ranked statements. We used a one-sample t-test to determine if the mean rank of statements

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was significantly higher or lower than the neutral score of zero. We also used the Mann-Whitney U test to determine if rankings differed based on gender (male, female) and age (< or >41 years old). This created the opportunity to explore how different demographics may prioritize impacts from climate change differently. The results from these analyses were shared with each community to get their interpretation of different perspectives and the overall results.

Community Comparison:

To compare between communities, we developed codes describing the statements derived from focus groups as described above. Three coders developed 17 codes (see codebook in Additional File 1 Supplemental Table 1), and we assigned up to four codes to each statement, as many statements touched on multiple topics. After discussions and iterative coding, we calculated an interrater reliability kappa to achieve a minimum 80% agreement across statements, as determined by Cohen's kappa statistic (McHugh 2012). Codes and their definitions are provided in the supplemental materials. For analysis codes that were assigned to statements from focus groups and interviews by at least 2 out of 3 reviewers were retained. Assignment of codes to all statements and the top 5 ranked priority statements was determined for all communities combined and compared between communities to identify possible differences between communities.

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Results

Demographics

We engaged a total of 324 adult residents from our four partner communities in ranking statements. Participants were 55% women and 45% men. Figure 2 shows the

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distribution was proportional across the six general age groups, with the exception of the 71 years and older group, which was smaller.

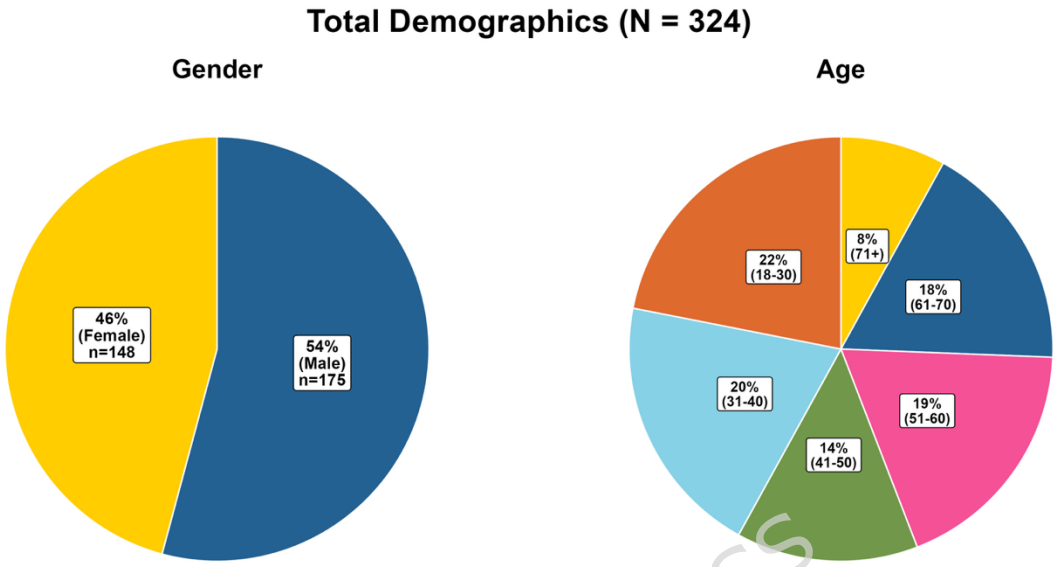


Figure **2:** Demographics of community members participating in ranking of priorities.

Community-level results from Q-methodology

Arctic Village

In Vashrąįį K'ǫǫ (Arctic Village), we conducted semi-structured individual and group interviews with 17 Gwich'in knowledge holders to gather formative data on perceptions and concerns about climate change effects on health and well-being in the community. Individual interviews were conducted based on community input to capture the priorities of knowledge holders who could not participate in the focus groups. Comments included reports of observed landscape change (Figure 3), changes in subsistence resources, i.e., animals, plants, and water, and the implications for holistic human health and cultural continuity.

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During focus-group discussions, landscape changes were linked to increased subsistence challenges, such as difficulty navigating land and rivers due to erosion, collapsing structures, sedimentation affecting river channels, water contamination, and a decline in animal habitats. Permafrost thaw also threatens village infrastructure and key cultural sites. Challenges to transferring Indigenous knowledge based on these disruptions were also described.[1] Participants suggested a need for monitoring and safety testing of water and subsistence foods, as well as raising awareness within and beyond the community to address uncertainty around the effects of climate change on the health and well-being of the community and its members. One Tribal Council member stated:

> *"When things change so much, we are not able to maybe show the young people the generational knowledge we have been building up over thousands of years, because it is no longer there to be practiced. Then, that being a really important part of our culture is affecting our mental health. Because here is that detachment that is much deeper than just food and a hungry stomach, even a hungry soul."*

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[1]Community members typically used terms such as traditional or generational knowledge to describe this. Here we have instead adopted the term Indigenous knowledge to reflect its current usage in the literature.

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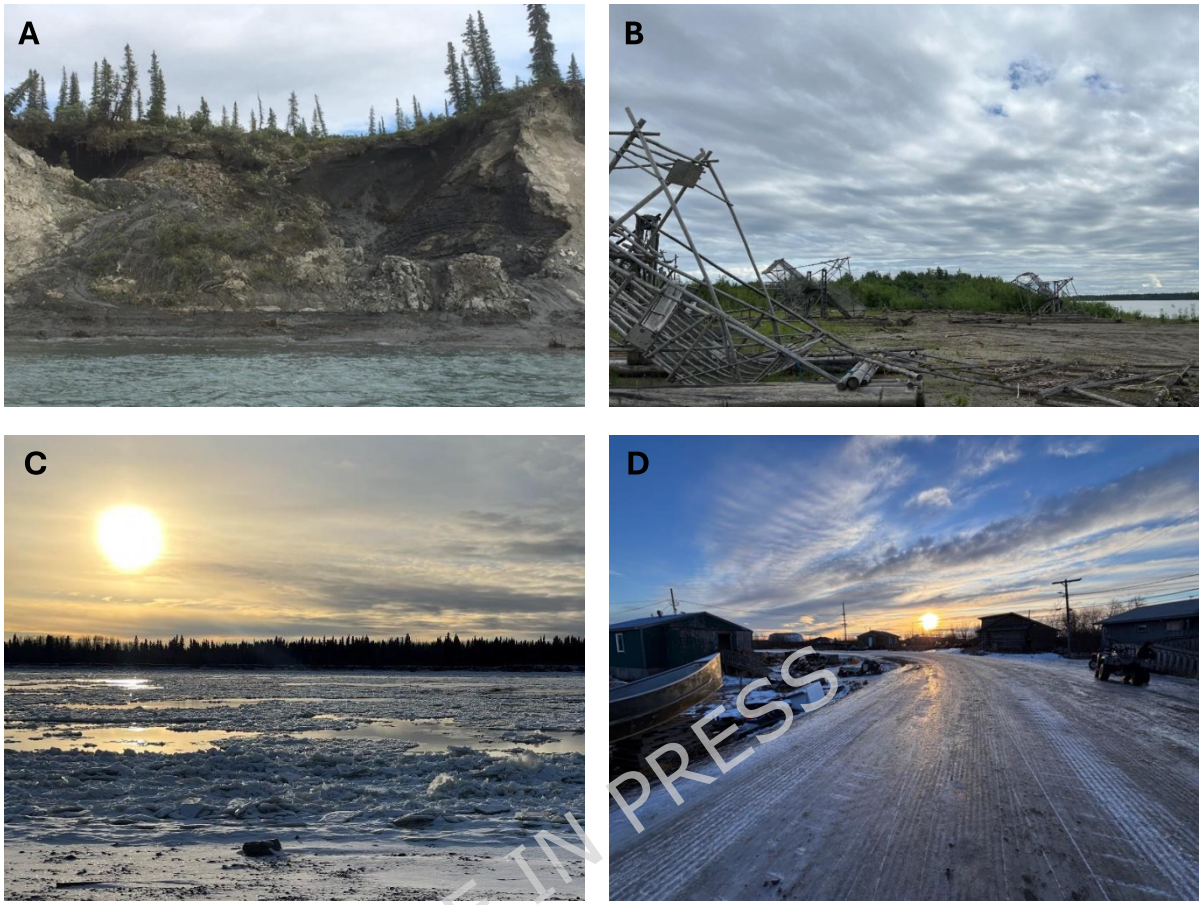


Figure 3: A: Evidence of erosion and permafrost thaw upriver from Vashrąįį K'ǫǫ on the K'aii'eechu'njik River. B: Fish wheels in Tanana going unused during Chinook and chum salmon migrations due to restrictions on fishing. C: Yukon River ice in Tanana observed to continue to flow on November 5th (a departure from historical conditions of complete freeze up prior to this date reported by community participants). D: Icy roads in Emmonak in December due to freezing rain.

Following focus groups, data were used to condense discussion topics to generate 26 clear statements of community concerns about climate change and health. All adult community members were invited to rank the statements using the Q-sort method individually. A total of 22 ranked datasets were collected. Table 1 shows the statements

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ordered by overall rank order. It is critical to note that all statements are important to community members, as indicated by the qualitative data, and that many statements are connected to a holistic understanding of the health impacts of climate change.

Table 1. Arctic Village community-derived statements about climate change and health are ranked in order. Mean statement scores (higher score = more important) for all participants, and differences in rankings by gender (female, male) and age (< or >41 years old) of participants, within Arctic Village. Statements included with a mean rank that was significantly (***p<0.001, **p>0.001 to ≤0.01, *p>0.01 to ≤0.05) greater than the neutral value of zero based on a one-sample t-test. The Mann-Whitney U test was used to determine if rankings differed by gender or age. If significant differences were estimated, we indicated which group had a greater median value.

Community = Arctic Village, N=22				
Rank	**Statements**	**Mean Score**	**Gender**	**Age**
1	Gwich'in knowledge about the land and animals becoming harder to pass on to the youth	1.7***	No	No
2	Behavior and abundance of animals like caribou, ducks, and fish is changing	1.6***	No	No
3	Concern about how climate change is threatening the health of animals like caribou and fish	1.4***	No	No
4	Concern about animal diseases threatening the safety of eating our traditional foods	1.1***	No	No
5	Drinking water quality threats from permafrost thaw	1.0**	No	No
6	Affects to physical health from lack of access to traditional foods	.8*	No	No
7	Changes to water depth and river channels making it difficult to navigate	.7	No	No
8	Abnormal and unpredictable weather that affects animals' access to food	.5	No	No
9	Permafrost thaw changing the land - walking trails no longer available, accessible, and/or safe	.4	No	No
10	Affects to mental health from lack of access to traditional foods	.3	No	No
11	Climate change negatively affecting mental and emotional health and well-being	.3	No	No
12	Abnormal and unpredictable weather that affects animal and human safety	.2	No	No

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13	Historical and cultural places threatened by permafrost thaw	.2	No	(younger> older)
14	Increasing temperatures and warmer water temperatures	.1	No	No
15	Inability to collect clean, fresh glacier water	.0	No	No
16	Glaciers melting and disappearing	.0	No	No
17	Challenges physically accessing traditional foods because of climate change	-.5	No	No
18	Concerns about changing balance and new plants and animals	-.5	No	No
19	Inability to collect clean, fresh glacier water	-.5	No	No
20	Increase intensity in lightning strikes affecting land and villages	-.6	No	No
21	Injustice of facing the impacts of climate change when we aren't creating it	-1.0**	No	No
22	Bears and wolves coming into the village more and threatening community safety	-1.0*	(male> female)	No
23	Cannot depend on frozen ground and glaciers for food preservation now	-1.2**	No	No
24	More bears and wolves killing caribou	-1.4***	No	No
25	Air pollution due to wildfires	-1.5***	No	No
26	Increase intensity of lightning strikes affecting land and villages	-2.0***	No	No

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Notably, many of the statements relate to a subsistence lifestyle. Sixteen directly reference access to healthy subsistence resources, including nine of the ten statements ranked most important by the community.

Tanana

During our visit to Tanana, we conducted focus groups and interviews with 43 community members. Many comments centered around the inability to practice cultural lifestyles and the impact on knowledge transfer and cultural experiences of young people in the community. Lack of access to fishing for salmon was mentioned frequently (Figure 3B), along with the economic and cultural impacts because of the inability to

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fish. Multiple community members also highlighted changes in winter weather and the late date at which the river would freeze and become safer for travel (Figure 3C). One focus group participant stated:

> *"Climate change is going to be a disaster for our livelihood. (We will see) impacts in every part of our life."*

Following focus groups, data were collated and condensed to generate 31 statements representing community concerns. All adult community members were then invited to rank the statements individually. A total of 54 ranked datasets were collected, which is approximately 23% of the adult population of Tanana. Table 2 shows the statements ordered by overall rank order.Data were analyzed to identify top priorities as well as patterns in the ranking of statements.

Table 2. Tanana community-derived statements about climate change and health in ranked order. Mean statement scores (higher score = more important) for all participants, and differences in rankings by gender (female, male) and age (< or >41 years old) of participants within Tanana. Statements included with a mean rank that was significantly (***p<0.001, **p>0.001 to ≤0.01, *p>0.01 to ≤0.05) greater than the neutral value of zero based on a one-sample t-test. The Mann-Whitney U test was used to determine if rankings differed by gender or age. If significant differences were estimated, we indicated which group had a greater median value.

Community = Tanana, N=54				
Rank	**Statements**	**Mean Score**	**Gender**	**Age**
1	Loss of ability to fish for salmon impacting overall individual and community wellbeing	2.9***	No	No
2	Cultural knowledge and experience not being transferred to youth	2.0***	No	No

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3	Seasonal weather changes impacting traditional hunting timing for moose and hunting regulations not aligning with the changing behavior.	1.6***	(female>male)	No
4	Inability to practice cultural lifestyles through activities like hunting and fishing, leading to food insecurity (storing enough food).	1.3**	No	No
5	More threats (warming, overfishing etc.) impacting fish health (small fish, disease) and numbers	1.1**	No	No
6	Lack of fish and additional financial burden leading to loss of dog mushing lifestyle	0.9**	No	No
7	Less cultural foods for sharing and to support ceremonies and other community events	0.9**	No	No
8	Less access to nutritional traditional foods creating greater reliance on storebought foods that is less healthy and more expensive	0.8*	No	No
9	More predators around community and in town because of a lack of natural food sources they rely on	0.8*	No	No
10	Impacts on physical, emotional, and mental health and purpose from inability to harvest and gather traditional foods	0.5	No	No
11	Local traditional knowledge not being "heard" by decision (policy) makers and agencies	0.4	No	No
12	Higher cost (time and money) to travel further to harvest animals	0.4	No	No
13	Changes in climate that make it dangerous to travel to get to traditional spring hunting areas (birds, trapping).	0.4	No	No
14	Changes in weather systems changing known/understood patterns migration and animal behaviors	0.1	No	No
15	New and more parasites and diseases attacking plants & animals (spruce bark beetle, white spots in fish, bird flu, etc.)	-0.0	No	No
16	Severe mental, emotional, and behavioral health impacts from trauma of rapid loss of hunting and fishing abilities.	-0.1	No	No
17	Later freezing (lakes, sloughs, river) impacting winter travel to camps, traplines, and to train dogs	-0.1	No	No
18	Extreme weather events (flooding, heavy rains, icing, high winds, storms) causing community stress and unsafe travel	-0.1	No	No
19	Reduced physical activity and health because less reason to get out on the land (no fishing, slow freeze up delaying sledding, etc.)	-0.3	No	No
20	Wildfire effects on people, animals, land.	-0.4	No	No
21	Heavy rains leading to landslides and bank erosion that change river conditions (fishing locations, navigation etc.)	-0.5	No	No
22	Unpredictable weather affecting berry patches	-0.9*	No	No
23	Increased erosion to banks, roads, and trails causing landslides impacting travel, trails, and property	-1.0**	No	No
24	Increase in rain freezing during the winter making it more dangerous to walk outside	-1.1**	(male>female)	No
25	More standing water making it more difficult to access customary hunting locations and trails	-1.1**	No	No

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26	Invasive plants (like vetch, chokecherry) that are harmful to native plants and animals moving into town	-1.2***	No	No
27	Overgrowth of plants impacting navigation in lakes, creeks, and sloughs	-1.2***	No	No
28	Changes in permafrost impacting buildings	-1.2**	No	No
29	Overgrowth of shrubs and young trees crowding out berries and blocking trail access	-1.4***	No	No
30	Increased risks to health from river/creek water (giardia/beaver fever etc.)	-1.5***	No	No
31	Less large wood in river for firewood	-2.0***	No	No

Similar to the results from Arctic Village, many of the most highly ranked concerns address subsistence activities, Indigenous knowledge, and how this knowledge is passed on to the next generation. The community highlighted the importance of fish to community well-being in ranking statements.

Toksook Bay

In Toksook Bay, we conducted multiple focus groups, including one with Elders conducted in Yup'ik, the first language of many Elders. Statements and observations related to how the older generations find it more challenging to reach out and teach younger people due to changing weather and changing ways of life.

The other focus groups were conducted in English. Participants in these groups focused on the impacts on the community's relationship to fish, noting how climate change affects access to all species that were historically harvested. The weather patterns and seasonal changes in tundra vegetation were once reliable indicators of when to begin fishing for different species. Today, these rhythms with the fish migration timing have shifted and are less predictable. The weather becoming wetter during

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fishing seasons also affects the ability to properly dry fish, causing food safety and, ultimately, health concerns. They also noted impacts on the berries, which are additional critical sources of cultural and physical sustenance.

We again condensed discussion topics to generate 30 statements of community concerns and conducted the sorting described above. A total of 113 ranked datasets were collected in Toksook Bay; Table 3 shows the statements ordered by overall rank order.

Table 3. Toksook Bay community-derived statements about climate change and health in ranked order. Mean statement scores (higher score = more important) for all participants, and differences in rankings by gender (female, male) and age (< or >41 years old) of participants, within Toksook Bay. Statements included with a mean rank that was significantly (***p<0.001, **p>0.001 to ≤0.01, *p>0.01 to ≤0.05) greater than the neutral value of zero based on a one-sample t-test. The Mann-Whitney U test was used to determine if rankings differed by gender or age. If significant differences were estimated, we indicated which group had a greater median value.

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Community = Toksook Bay, N=113					
Rank	**Statements**	**Mean Score**	**Gender**		**Age**
1	Concern for young people being able to practice subsistence lifestyle in future	1.3***	No		No
2	Peoples physical health not being as good from not harvesting and eating as many Yup'ik foods as before	0.6**	No		No
3	Increase in respiratory health issues, such as pneumonia, coughs and common colds lasting longer	0.6*	No		No
4	Increase in diagnosis of cancer, heart attack, and stroke	0.5*	No		No
5	Erosion is getting worse because water level is higher	0.5*	No		No
6	Unpredictable weather making it harder to preserve food through drying	0.5*	(female>male)	No	
7	Increase in signs of infection and unusual growths in fish, land and marine mammals	0.5*	No		No
8	Not able to put away the amount of fish needed for families	0.5*	(female>male)	No	

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9	Erosion causing the release of dump sites and other pollution into the water	0.4	No	No
10	Permafrost melt and erosion impacting the community	0.4	No	No
11	Less fish in the rivers and the bay	0.4	No	No
12	Weather causing changes in fish habitats and migration of fish (when they arrive or leave)	0.3	No	No
13	Increase in flooding and higher waters	0.3	Yes	No
14	Berry picking sites are changing	0.2	No	No
15	The ice is not forming fully, affecting sea mammals and travel for people	0.1	No	No
16	More storms with increased intensity	0.0	No	No
17	Unpredictable weather making it harder to know when to go out to harvest fish and animals	0.0	No	No
18	Increase in invasive species, such as beavers, which are impacting fish migration and water quality	0.0	No	No
19	More asthma and allergies in young people	0.0	No	No
20	Water channels and landmarks are changing, which effects travel for hunting and fishing	-0.1	No	No
21	Decrease in quality of land animals, such as muskox and moose	-0.2	No	No
22	Increased regularity of high winds	-0.3	No	No
23	Less snow cover effecting the plants and ability to travel in winter	-0.4	No	No
24	Tundra is changing, becoming wetter or dryer in different places. which is affecting the geoblock trails and plants	-0.4*	No	No
25	Fewer shore birds returning to the village and some are birds disappearing	-0.5*	No	No
26	Reduced size of fish	-0.6**	No	No
27	Tundra plants not blooming when they should be	-0.7**	No	No
28	Water and river weeds growth showing up and makes some waterways impassable	-0.9***	No	No
29	Salt content due to flooding impacting water quality	-1.1***	No	No
30	Sadness from not being able to be on the land.	-1.8***	No	No

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While Toksook Bay is located in far western Alaska, similar to villages in the Interior (Tanana and Arctic Village), natural resources and subsistence featured prominently in the top-ranked concerns about climate and health. However, health concerns related to climate impacts on air quality, such as increased respiratory illnesses and chronic diseases, were also highly ranked.

Emmonak

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In Emmonak, we conducted focus groups and interviews with 47 individuals to inform the generation of statements for ranking. We again had one group of Elders where discussion was largely in Yup'ik. During our community visit, unseasonably warm weather led to very dangerous conditions that limited mobility in the community because of extremely icy roads (Figure 3D). Such travel difficulties caused by climate change were one of the issues raised in the focus groups, with this theme also extending to the dangerous conditions of river ice that freezes much later and is less solid and reliable for travel. Participants also emphasized the impacts of the changing rivers on the salmon migration up the Yukon River and the multiple factors leading to the loss of economic opportunities and Indigenous knowledge transmission related to salmon harvesting. More specific to this community, were statements that linked climate changes and changes in traditional ways of living to increased problems related to mental health, family well-being, and substance misuse.

Based on community comments, we developed 29 statements. A total of 134 ranked datasets were collected in Emmonak; Table 4 shows the statements ordered by overall rank order.

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Table 4. Emmonak community-derived statements about climate change and health in ranked order. Mean statement scores (higher score = more important) for all participants, and differences in rankings by gender (female, male) and age (< or >41 years old) of participants, within Emmonak. Statements included with a mean rank that was significantly (***p<0.001, **p>0.001 to ≤0.01, *p>0.01 to ≤0.05) greater than the neutral value of zero based on a one-sample t-test. Mann-Whitney U test was used to determine if rankings differed by gender or

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age. If significant differences were estimated, we indicated which group had a greater median value.

Community = Emmonak, N=134				
Rank	**Statements**	**Mean Score**	**Gender**	**Age**
1	Young people are not able to learn their traditional ways of life (cutting fish, going to camp, etc.) or how to keep safe on the land/water	1.7***	No	No
2	Loss of ability to fish for salmon impacting overall individual and community wellbeing	1.6***	No	No
3	More threats (warming, overfishing, etc.) impacting fish health (small fish, disease) and numbers	.9***	No	No
4	Inability to practice traditional way of life leading to more drug and alcohol use	.9***	No	No
5	Less access to traditional foods creating greater dependence on stores and food insecurity	.7**	No	No
6	Changes in climate that make it dangerous to travel	.5*	No	No
7	Young people not seeing a future for themselves in the community	.3	No	No
8	Families not going out to camp anymore	.3	No	No
9	Climate change leading to more domestic violence (fewer jobs, people staying home more)	.3	(female>male)	No
10	Climate change creating more dangerous conditions in communities (icy roads, more storms)	.2	No	No
11	Climate change impacts on sharing (food, resources)	.1	No	No
12	Increased erosion to river banks, roads, and trails	.1	No	No
13	More cancers and new illnesses (iron deficiency, coughing)	.1	No	No
14	Climate change is damaging subsistence equipment (snowmachine tracks, etc.)	.0	No	No
15	Reduced physical activity and health because less reason to get out on the land (no fishing)	.0	No	No
16	Changes in permafrost impacting fish camps and buildings	-.1	No	No
17	River channels (travel) are changing faster- need to keep relearning	-.1	No	No
18	Weather is more unpredictable	-.2	No	No
19	Permafrost melt damaging building and releasing toxins into the ground	-.3	No	No
20	Trash/dumps and its damage to ground water and environmental areas	-.3	No	No
21	Less access to medical care and medicines due to climate change	-.3	No	No
22	Spiritual connection to the fish and animals harder to maintain	-.4	(male>female)	No

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23	Climate change making existing problems worse	-.4*	No	No
24	Not being able to travel leading to social disconnection (families can't see each other)	-.5*	(male>female)	No
25	New and more parasites and diseases attacking animals	-.8***	No	No
26	Changes in weather systems changing animal behavior	-.9***	No	No
27	New animals (beavers) damaging waterways and impacting fishing	-.9***	No	No
28	Climate change causing people to leave the community for jobs	-1.2***	No	No
29	Seawall is getting smaller and there are more floods	-1.4***	No	No

Similar to the other three sites, community members ranked climate impacts on Indigenous knowledge and subsistence resources highly, as well as the effects of climate-driven loss of economic opportunities on behavioral health indicators.

Cross-site community results from coding

To assess similarities and differences between communities, we assigned codes to statements from each community. We assigned up to four codes to each statement, as many statements touched on multiple topics. When comparing the codes assigned to all statements, the most frequently assigned codes are listed in Table 5 and Figure 4. All codes are shown in Additional File 2 Figures S1-S4.

Table 5. Most frequently assigned codes in all communities with the number of statements they were assigned to.

Code	# assigned
1. Lack of Traditional Foods / Food Security	29
2. Travel	28
3. Changes to Plants/Animal Populations and Behaviors	27
4. Personal/Community Safety Concerns	23
5. Changes in Weather Conditions (including Extreme Weather Events)	23

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Most codes are assigned to statements from all communities with three exceptions: Economic Impact, Policy and Justice, and Access to Services. The code of Economic Impact was assigned to statements from only two of the four communities: Emmonak and Tanana. These two communities are both located on the Yukon River, which traditionally has had a thriving economic base with the Yukon salmon fisheries that have, in recent years, been greatly reduced, necessitating fishery closures mandated by management agencies for commercial and, even more recently, subsistence purposes. Similarly, policy and justice were assigned to statements from the two communities in Interior Alaska, where self-governance is strongly centered in the Tribal structures.

Cultural and contextual factors may influence the generation of priorities, as indicated by results from the total codes applied, but when we focused the analysis on the top five highest-ranked priority statements from each community, the most commonly assigned codes to this subset of statements more closely aligned and are listed in Table 6 and Figure 4.

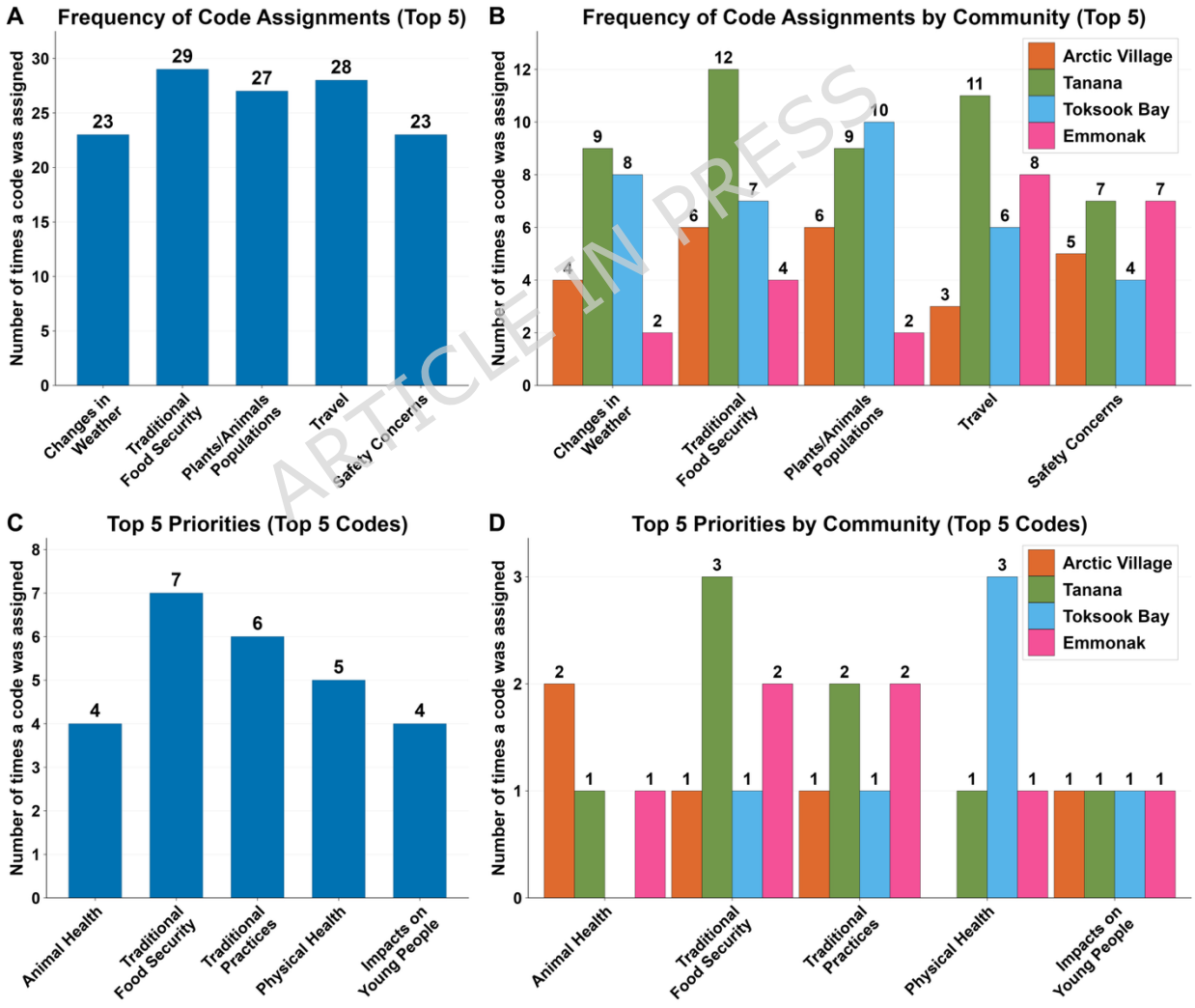
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Table 6. Codes assigned to top five ranked statements in each community, and the number of statements among the top 5 a code was assigned to.

Code	# assigned in top 5
1. Lack of Traditional Foods / Food Security	7
2. Loss of Traditional and Cultural Practices (including knowledge transfer)	6
3. Physical Health	5
4. Impacts on Young People	4
5. Animal Health	4

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Three codes (Lack of Traditional Foods / Food Security, Loss of Traditional and Cultural Practices (including knowledge transfer), Impacts on Young People were assigned to the five highest-ranked priority statements in all communities. Three additional codes (Animal Health, Physical Health, Traditional Food Quality and Safety) were assigned to the five highest-ranked priority statements in three out of the four communities. Overall, these findings suggest shared challenges and priorities between communities, especially in the highly ranked areas of traditional food and cultural practices and climate change impacts on young people.



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Figure 4: Comparison of priorities between communities. Statements were analyzed by how often codes were assigned for all statements (A&B) as well as for the five top-ranked statements for each community (C&D). The 5 most frequently assigned codes are shown. Figures with all codes are provided as supplemental figures 1-4. Results are presented for all communities combined (A&C) and disaggregated by community (B&D).

Discussion

In this study, we examined how AN communities prioritize their concerns regarding the various current impacts of climate change on their health. The study sampled across four small, geographically remote AN communities broadly representative of two diverse rural regions in Interior and Western Alaska. Our primary finding is that the four participating AN communities prioritized climate change impacts on access to traditional foods, loss of traditional and cultural practices, including the ability to pass on Indigenous knowledge to young people, and young people's well-being as their most significant concerns. These priorities aligned across these communities and, in general, across the demographic variables of gender and age, and the two regions. Additionally, the impacts of climate change on the health of animal subsistence resources, human physical health, and traditional food quality and safety were each prioritized in three out of the four communities, and concerns associated with impacts on community physical infrastructure and general well-being were also frequently prioritized.

Discussions at our alliance meetings and during community visits clarified a central theme underlying their prioritizations: health is understood by community members through a broader, interrelated, and more holistic perspective than is typical among

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many Western medical providers, agencies, and scientists. This perspective aligns with other Indigenous definitions of health (Donatuto et al. 2011; Donatuto et al. 2016, 2020). Any relevant concept of climate change impacts on health in these rural AN settings requires consideration of health and the overall well-being of entire communities. For example, in the community of Tanana, while lack of fish and fishing were discussed in the context of nutrition, physical health, and food security, a strong emphasis was also placed on the loss of cultural practices like summer fish camps, where families participated together in traditional activities and knowledge was transferred. Community is bounded in this understanding not merely by its human inhabitants and includes the important human connections to place encompassing the health of the animals and plants in their surroundings, which constitute the foundation of Indigenous ways of living and thriving in rural Alaska.

While such expanded understanding may reveal some of the broader connections underlying the impacts of climate change on AN health to include the interrelated impact upon the land, its animals, and people, it also introduces complexity, making simple identification of priorities more difficult, as all aspects of human health, the environment, and the community are interconnected. Such difficulty is more than counterbalanced through the enhancement in our ability to provide culturally relevant approaches to health research, including an intervention science that can develop effective strategies that fit this culturally distinct context to facilitate an adaptation to climate change through alignment with community priorities that reflect cultural paradigms of health.

Concerns with fish and animal health, and with air and water quality, were often stated through community questions, and the community identified priorities for greater

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community access (Cold et al. 2020) and capabilities to conduct scientific testing of water, air, fish, and mammals to determine their level of health and potential contamination. Communities prioritized having more information about their environment, including access to greater Western scientific knowledge provided from scientists who come frequently to communities and, for example, conduct testing on fisheries or sampling from the permafrost and waters. Findings from this study indicate an interest and need to integrate Western science-driven knowledge and practices into Indigenous culturally-grounded solutions that address climate and health impacts. The potential alignments across Indigenous holistic and Western One Health perspectives offer one promising avenue in guiding these efforts (Heeringa et al. 2019).

For example, while many community-identified priorities were associated with the health, availability, and accessibility of subsistence resources, as well as the transmission and preservation of Indigenous knowledge and ways of life, health and safety of animal resources, threats to community infrastructure, and economic opportunities were also linked to the health and well-being of individuals and communities. Together, these priorities and the approach to understanding health underlying them suggest alignments with the One Health paradigm, with a particular relevance to Northern communities (Sustainable Development Working Group 2017; Hueffer et al. 2019; Reynolds et al. 2022). The connection between the land and the peoples of the North, especially among Indigenous peoples, is more intimate than in many other geographical and more urban regions. One priority, food security through the availability of resources such as animals, berries, and other plants, also aligns with previous findings (Flint et al. 2011; Unc et al. 2025). Given the high cost of fuel and

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many food items (Brinkman et al. 2014; Tiff-Annie et al. 2025), especially in rural Alaska, access to subsistence activities retains importance in the North, where food insecurity is increasing (Bjerregaard 2001). But the availability of subsistence resources is important to well-being far beyond providing nutrients, linking food security to the survivance of cultural practices. Subsistence has importance not merely for sustenance but also for cultural, emotional, and spiritual well-being (Flint et al. 2011; Chatwood et al. 2017). The subsistence life way's aligned connection to land has been identified as an important protective factor in resilience processes, especially in Indigenous communities in the North (Rasmus, Charles, et al. 2014; Rasmus et al. 2019). In this way, the findings of this study are consistent with both the One Health perspective and a holistic view of health that includes persistence and survivance of traditional practices, as exemplified in the above quote from a council member invoking, "not just a hungry stomach but also a hungry soul. Climate stressors cannot be sufficiently addressed without accounting for both the influence of cultural factors in AN community understandings of their impact, as well as their impact upon the culture itself (Haven et al. 2025).

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Finally, effects of climate change on mental health in the circumpolar North will likely be dramatic but are currently understudied and poorly understood (Willox et al. 2015; Lebel et al. 2022). Willox and colleagues proposed five main pathways by which climate change can influence mental health: through change in sense of place and traditional practices that promote well-being; through effects on physical health; through deterioration of built infrastructure and resulting stress; through negative narratives of climate change ("disaster narratives"); and through other compounding stressors such

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as colonization, acculturation, and low economic status. Connection to land and sense of place and culture are important resilience factors in the North, especially for Indigenous youth (Macdonald et al. 2013; Allen et al. 2014; Rasmus et al. 2019).

The main limitation of this study is the possibility that a subset of community members who participated in both the focus groups and the ranking had opinions on the impact of climate change on health that are systematically different from the community overall. However, our high participation rate and follow-up conversation with tribal leadership and other community representatives make this unlikely.

Conclusions

A driving purpose for this study was to identify the climate and health priorities of AN communities to guide the implementation of accessible, culturally relevant solutions to climate change adaptation and the assessment of their effectiveness. All of the communities' highly ranked priorities related to the transmission of Indigenous knowledge and practices to the younger generations. Additionally, all communities prioritized the continuity of subsistence practices and the health and accessibility of subsistence resources. These priorities indicate the importance of climate and health solutions that increase opportunities for intergenerational knowledge transmission and subsistence engagement in the everyday context of rural AN communities. Results from this study confirm the need and importance of culturally grounded interventions that promote holistic health in rural communities. Our findings align with our long-term community-based participatory research initiative in western Alaska that has demonstrated a causal relationship between engagement of young people in their

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traditional Indigenous ways of life and subsistence practices and increased reasons for life and well-being (Rasmus et al. 2019; Allen et al. 2023).

The importance of continued community engagement to understanding and enacting climate and health solutions in AN communities was most eloquently stated by an Elder in one of our focus groups:

> *"Tailuci, wavet aqumluci, calillgutekluta, apqaurluta, kangiiturluki ciin makut pugat ellamta cimirturallra maliggluku? Irniamta atu'urkaitnek calilliniuten, naaqikuneng eliciiqut, nalluyagucuitut wangkucicetun." ~ "Come here, sit down with us, ask questions, learn the why in these changes. Share the knowledge, I see that you are working on something that our children will use for themselves. When they read about it, they will learn, they don't forget things they've learned just like us (adults)."*

Our team recognizes and honors this approach through community-directed alliances that emphasize community worldviews and their aligned priorities.

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List of Abbreviations

AN: Alaska Native.

Declarations

• Ethics approval and consent to participate:

This work was reviewed by the University of Alaska IRB (protocol 2118090) and the Yukon Kuskokwim Health Corporation's Executive Board of Directors (proposal 24.06.10) and followed principles of the Helsinki Declaration and the Supplemental Information to the NIH Policy for Data Management and Sharing: Responsible

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Management and Sharing of American Indian/Alaska Native Participant Data. All participants provided informed consent before participating in this study,

• Consent for publication: Not applicable

• Data Availability: The manuscript's data will not be deposited.

• Competing interests: The authors declare that they have no competing interests.

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• Authors' contributions: K.H and S.R conceived the research; K.H., T.B., H.R., B.C., V.J., J.A. C., S.J., M.C., J.S., T.P., A.B., S.R. conducted the research; K.H., T.B., V.J., S.R. analyzed the results; K.H., T.B., H.R., M.C., J.S., T.P., A.B., J.A., R.D.A., A.G.-K., K.S.L., S.R. wrote the paper. All authors read and approved the final manuscript

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Additional files:

Additional File 1: Supplemental Table 1

Supplemental Table 1: Codebook with shortened code names and Code definitions.

Additional File 2: Figures S1-S4

Figure S1: Comparison of priorities between communities. Statements were analyzed by how often codes were assigned for all statements for all communities combined.

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Figure S2: Comparison of priorities between communities. Statements were analyzed by how often codes were assigned for all statements disaggregated by community.

Figure S3: Comparison of priorities between communities. Statements were analyzed by how often codes were assigned for the five top-ranked statements for each community for all communities combined.

Figure S4: Comparison of priorities between communities. Statements were analyzed by how often codes were assigned to the five top-ranked statements for each community, disaggregated by community.

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